

Module 13: Dividend Policy

Topic 229: Concept of dividend

We are starting our discussion on Dividend Policy but before going into the detail we first have to see that what is Dividend.

Dividend:

Dividend

- Profit distributed among shareholders
- Paid out of Retained Earnings
- Annual dividend
- Interim dividend



Profit distributed among shareholders:

Dividend is actually the distribution of profit as whenever any business earns profit and distributes this profit among its owners or shareholders. So, the profit which is distributed among shareholders is termed as **Dividend**.

All of the profit may be distributed among the shareholders or a part of the profit may be distributed or in some years, the profit might be earned but nothing is distributed among the shareholders so, the distribution of dividend is not certain as there may be dividend or may not be dividend. So, there are all possibilities of having dividend.

Paid out of Retained Earnings:

Dividend is paid from the profit but this profit is firstly added in our accumulated profit or retained earning. So, for the payment of Dividend, we need not to have current year's profit meaning that in such a year in which even any company has not earned any profit or has gone in loss and also in that year the Dividend can be paid out. Because the dividend is not paid out of the current year's profit but it is paid out of the accumulated profits which we name **Retained Earnings**. If we have Retained Earnings or Retained Profits are present then we can pay our dividend from that profit.

Annual Dividend:

The Dividend is usually paid at the end of the year when the amount of profit has been calculated.

Interim Dividend:

But if there is a surety that the company is doing so well that it is likely to have a lot of profit by the end of the year then this company also pays Dividend during the year and that is called **Interim Dividend**.

Why should a company pay Dividend?

- Return of investor
 - Dividend
 - Capital Gain
- Common share dividend
- Preferred share dividend



Because Dividend is the return of the investor or the shareholder meaning that shareholder has invested his money in any business for the reason that to earn return and that return is received in two forms as:

Dividend: The Dividend that a company pays him.

Capital Gain: The Capital Gain which he/she get through purchase or sale of shares. Capital Gain simply means that if we purchase an asset and sell it and if we get any profit or return from the difference of that buying or selling price then it is called Capital Gain. When we buy a stock on lesser price and sell a stock at higher price then the difference in the two is the Capital Gain of the investor.

So, the investor is not only looking for Capital Gain but investor is also looking for the profit which is there with the company. These two things collectively make the return of an investor. As investor goes towards Capital Gain but he also wants to get dividend also.

The Capital Gain can be earned if we frequently buy and sell shares but dividend cannot be earned by quickly buying and selling shares as to earn dividend, we have to hold the stock or shares for a certain period of time. As we have said that generally companies give annual dividend and if we purchase a shares then theoretically speaking we have to hold that shares with one year in order to get dividend on that share.

Another important thing that we have to keep in mind that is related to Dividend or Capital Gain which is; investment in Shares/Securities/Stock Exchanges is not gambling, it is not speculation, it is about earning normal return over longer period of time because Stock Exchanges are called Capital Markets and in Capital Markets, the investment never for short term period of time.

Common Share Dividend and Preferred Share Dividend: Dividend is paid for the reasons because it is return for the shareholder or investors and they have made this investment for getting that return. This dividend is paid on both types of shares meaning that we have Common stocks and also have Preferred stocks.

Common shares or stocks are those which represent the true ownership of any company and they have an element of risk more as compare to the Preferred shares. The dividend is paid on these Common shares after the payment of dividend to Preferred shareholder. So, Common stocks holders may or may not get their dividend if it is all distributed among Preferred shares.

And **Preferred share dividend** is actually a fixed percentage of profit that we get.

We can conclude our discussion that Dividend is essential because it is requirement of an investor and this dividend along with Capital Gain makes the total return of the investor. Common shares and Preferred shares both have got a certain dividend.

Module 13: Dividend Policy

Topic 230: Dividend policy

In previous lecture, we have talked about what is Dividend and what are various types of Dividends.

Now in this lecture, we will talk about the Dividend Policy or the Dividend Policies but even before that let's see that what a Policy is.

Policy:

Policy:

A system of principles to guide decisions and achieve rational outcomes



A **Policy** is a system of principles that guide decisions to achieve some rational outcomes. So, in every organization we have got many policies; policies related to management, marketing, human resource, production etc.

When we talk about the policy related to Dividends or the payment of Dividends then we call it a **Dividend Policy.**

Dividend Policy:

Dividend Policy has several aspects meaning that in Dividend Policy, there may be that if we have to pay dividend or not, company we make decision every year by considering some factors that we have to pay dividend this year or not.

Similarly a company may have to decide as a part of Dividend Policy that they would always pay Dividend but that amount of Dividend or percentage of Dividend may vary over time. So, all these aspects may be the part of Dividend Policy.



The structure of dividend payouts or retention of profit:

When we say that what is the crux of Dividend Policy then we can say that **it is the structure of dividend payouts** where **dividend payouts** means that how much will be paid to the shareholders in the form of dividend from our earnings. As we have discussed about **DPO (Dividend Payout) ratio** which we get when we divide dividend per share with the earning per share then we get a percentage of dividend.

It means that **Dividend Policy** includes determining that percentage of dividend that there will be how much percent of dividend payout. The remaining Dividend Payout is named to be Retention or Plough-back.

When we pay dividend as a business or company then we also retain a part of profit and that part of profit which is not distributed among the shareholders as

dividend is called the **Retained Profit** and this is calculated by considering the Retention Ratio or Plough-back ratio.

So, this is also that part of Dividend Policy that there will be how much percent of Dividend Payout and how much percent will be Retention.

Dividend Policies:

We have **two types of policies** related to the payment of dividends;

Irrelevance Policies of Dividend: It means that some people think that there is no difference whether to pay dividend or not meaning that an investor is indifferent whether he gets any dividend or not.

The **reason** is that as if investor takes dividend, its share price gets decrease and there will be no difference in his wealth and if he has not given dividend then his wealth goes at a higher level.

The proponents of Irrelevance theory says that it does not make any difference whether an investor gets dividend or he does not get dividend.

Relevance Policies of Dividend: On the other side, we also have another school of thought and that is the Relevance of Dividend meaning that these are those people who says that it is not right that there is no difference whether to pay dividend or not.

Both the advocates of Irrelevance of dividend and Relevance of dividend actually have their own set of arguments, logics, own way of looking at the decision that we will see in the next lectures.

Module 13: Dividend Policy

Topic 231: Dividend payout and retention

We have talked about the Dividend and we have also discussed that there are two policies which are totally different from each other with reference to payment of Dividend which we call them Relevance of dividend and Irrelevance of dividend policies.

To understand these policies, we need to understand the Dividend Payout and Retention and the impact of Retention and Payout.

Dividend Payout:

It is the amount of profit which is paid as dividend and in terms of ratio, it is calculated as the percentage of profit which is distributed as dividend. This ratio can be calculated by considering the total amount of dividend and total amount of profit or it may be calculated by considering the dividend of one share and the profit of one share.

Dividend Payout Formula:

So, the Dividend Payout ratio is calculated as:

$$\text{DPO (Dividend Payout) ratio} = \text{DPS/EPS}$$

Where:

DPO = Dividend Payout

DPS = Dividend per Share

EPS = Earnings per Share

Retention:

We have discussed a bit that whatever distributed is dividend then we also have some amount of profit which is not distributed so, the profit which is not distributed as dividend among the shareholders is the Retained Profit and that is called Retention.

So, if we have got a percentage of profit paid as dividend then we also have a percentage of profit that is retained that is called the **Retention Ratio**. It means that if we divide retained profit with our total profit then we get Retention Ratio.

Similarly if we calculate Retention per share and divide it with Earning per share then we can also get the percentage of Retention ratio. But generally we calculate Retention ratio that we also call **Plough back ratio** through Dividend Payout as given below:

Formula:

- **RR (Retention Ratio) = (1- DPS)/EPS**

Where:

RR = Retention Ratio

DPS = Dividend per Share

EPS = Earnings per Share

Effect of DPO (Dividend Payout ratio) and RR (Retention Ratio):

Effect of DPO and RR

- Higher DPO reduces growth
- Higher RR increases future growth



If the profit earned by a company or a business is distributed then there will be no change in the size of its Balance Sheet; suppose a company has started with 100 million, they earned 5 million profit and distributed the whole 5 million then after 5 years, this company will be with 100 million.

But if company retained that profit instead of distributing it as a whole then after one year this company will be with 105 million and in the next year or second year when they will earn profit, they will not earn profit on 100 million but they will earn profit on 105 million which will be definitely more than the profit of 1st year.

It is necessary to keep in mind that the payout of dividend is good because shareholders get happy with it as it is their return but if company wants to grow then it has to retain some profit.

So, we can say that **Higher DPO (Dividend Payout) ratio actually reduces growth because** we have distributed all the amount of profit that we can reinvest and can earn more profit.

But if we retain that amount of profit then we can earn more profit. So, we can say that **Higher Retention Ratio increases the future growth** meaning that any company's growth is related to that this company is adopting which dividend policy and how much percentage it has to pay dividend out of the earnings and the result is that more is that amount which is distributed, lesser is the growth.

Because growth depends on Retention as more will be our Retention, more will be the growth of the company against which if more will be dividend payout, more the growth of this company reduces or curtailed.

We can conclude our discussion that Dividend Policies are about Payouts and Retentions and these Payouts and Retentions have a direct impact on the future growth of the company so, the policies need to consider this element.

Module 13: Dividend Policy

Topic 232: Dividend Yield

We have talked about some of the basics of dividend, the payment of dividend, the amount of profit and retention.

In this lecture, we will be talking about the Yield or the Return to the investors through dividend.

Yield:



The term **Yield** is often used as the output that production that even for agriculture produce, we use the term Yield. But when Yield is used in stock markets with reference to the returns of the investors, this is actually the return of an investor with reference to its current market price.

The investor or a shareholder purchases a stock at a certain market price and that market price may not be the current price because if we are holding a stock let's say for 3 or 4 months, the price might have gone up or gone down and the return that we get today may be in the form of dividend or even if it is capital gain then we need to relate this return to the price which is existing today.

So, here the term **Yield** simply refers to the Return and this return is to be calculated with reference to the current market price of the investment. There are

different ways of calculating return of an investor but as we have said that if relate this return to the current existing market price that is called **Yield**.

Stock Yield:



The term **Stock Yield** is commonly used which means that a particular stock how much return it is providing or how much return to the shareholder a particular stock of the company gives that is the term most commonly used as the Stock Yield.

Likewise, if we are talking about bonds, the term would be **Bond Yield** or in stocks if we go into common and preferred shares or stocks then we may use the term **Common Stock Yield and Preferred Stock Yield**.

So, the **Stock Yield** is the total return of a shareholder with reference to the current market price and this will include the capital gain and the dividend which we have already discussed earlier that **any investor or shareholder gets two types of returns**; one return which they get from the difference of market price or difference of buying and selling price and the other return which they get in the form of dividend.

Dividend Yield:

In this discussion, we are referring to the Dividend Yield meaning that the dividend which is paid, the company relates that dividend declared generally to the

par value. It means that if any company will announce a dividend then it can say that this company has given 100% dividend.

And now, the question is that **this 100% is of what value?**

So, generally it is the relationship with the par value. Suppose a company's stock has a par value of Rs.10 and the company has paid dividend of Rs.10 so, relating Rs.10 dividend to the par value of Rs.10 then **it becomes 100%**.

But the question is that as an investor, if we have bought this share not for Rs.10 but let's say bought for Rs.100 then Rs.10 dividend with reference to a Rs.100 current market price then **it will only be 10%**. And this is the concept of **Dividend Yield**.

Dividend Yield

- Dividend is declared as %age of Par Value
- Dividend Yield is relating DPS to per share market price



So, **Dividend** is declared with reference to the par value but it is to be calculated with reference to the current market price of the stock.

We can conclude our discussion that **Yield** means the relationship of return to the current market price and it is a very fair way to calculate the return because if we have invested a certain amount today or if we sell this amount today then the amount that we will get for reinvestment purposes should be the amount which should be related with the return that we are getting.

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Topic 233: Passive and Active Dividend Policies

From previous lectures, we are building a base for the discussion of Dividend Policy and in this lecture we will be talking about the two contrasting **Dividend Policies** which are **Active and Passive Dividend Policies**.

The word **Passive** means that something which is not active, something which is not moving something, which is considered to be not with a particular objective and that variation exist in the other policy that is called **Active Dividend Policy** where we have got certain objectives, certain targets to meet and accordingly a policy is set.

Dividend Passive Residual or Passive Dividend Policy:

But with reference to Dividend Policy that is a **Passive Dividend Policy**, it also has a specific meaning and **we also call this the Dividend Passive Residual**. The word **Residual** means the leftover part and here when we say **Passive Residual** then we means the profit which is the residual profit or leftover profit which means such profit that we do not need in our business.

The question will be **why there is such profit which is not required in the business?**

The answer is that profit once earned by a company or by an organization can either be reinvested in the business or it can be distributed among the shareholders.

Now, if it has to be reinvested then there has to be reinvestment opportunity in the business meaning that we are running any textile firm or manufacturing any other product then the profit which we have earned, can be used for the expansion of our business or to develop the product line or diversify it and if we have no reinvestment opportunity then we can say that it is our Residual Profit.

The **Dividend Passive Policy** says that only such amount of profit will be given in the form of dividend which is leftover profit or whose reinvestment opportunity is not available in the business.

Here another important point is that **if this profit is distributed among shareholders then they can or cannot reinvest in the business?**

Definitely, they are investors and they have invested their money in this business and that is the point that if investment is made in our company or business it is because investor think that this is the best opportunity of investment for the money. And if we have that opportunity to reinvest their profit in our business then we should go for it and only if and only if this opportunity does not exist, the amount which is remaining should be distributed as dividend.

Dividend Passive Residual

- Irrelevance of Dividend
- Passive Residual refers to leftover profit (having no investment opportunity)



Question 1:

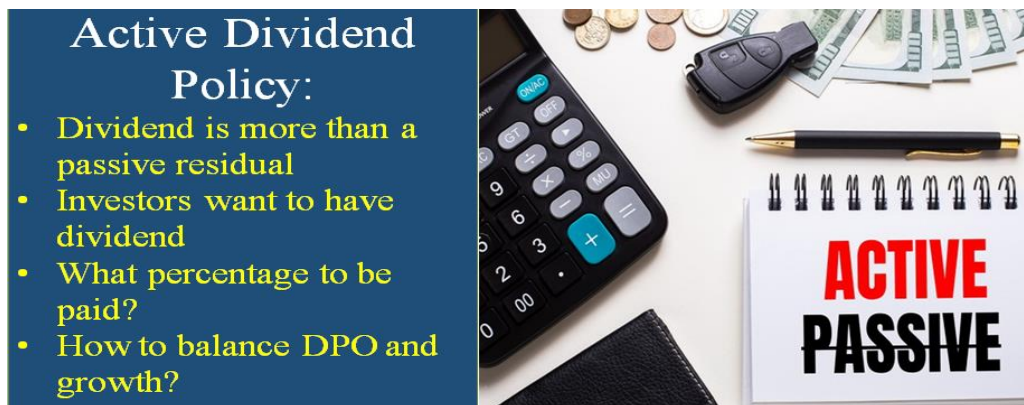
Should policy be active or passive?

So, this means that this **Dividend Passive Residual Policy** actually a policy of **Irrelevance of Dividend**. The Irrelevance Policy says that payout of dividend or not to pay dividend both is same for the investor, it is indifferent for investors.

Second thing is that **Passive Residual refers to leftover (having no investment opportunity)**. Here **Investment Opportunity** means that if that amount of profit which has to distribute among shareholders and if company can earn more profit by reinvesting this profit by itself in the business as compare to the shareholders that we call it **Investment Opportunity**.

Active Dividend Policy:

The first thing that Active Dividend Policy states that Dividend is more than a passive residual or Dividend is not just a passive residual so, the decision that whether we have an investment opportunity or not and then pay dividend is **not correct**. So, it is the negation of the first policy (Passive Dividend Policy).



The second is that Investors want to have dividend which we will elaborate in next lectures.

Then the question is that if we decide to pay dividend then **what percentage of profit should be paid as dividend?**

It should be continuously pay or fix percentage of the profit earned? But profit fluctuates as it is sometimes more or sometimes less then is the dividend paid according to the percentage of profit or it should be fixed? And this is a discussion of the stability dividend that we will discuss in next lectures as well.

How to balance DPO (Dividend Payout) ratio and growth of company?

Both DPO and growth of a company are important. In Active Dividend Policy, managers make decision by considering both Dividend payout ratio and growth that they have to pay how much dividend meaning that if preference of shareholders is dividend then there has to be dividend payout and simultaneously if the company wants to grow then it also has to retain certain amount of profit to reinvest in the business.

We can conclude our discussion that there are basically two policies related to Dividend are: Active Dividend Policy and Passive Dividend Policy. In **Passive**

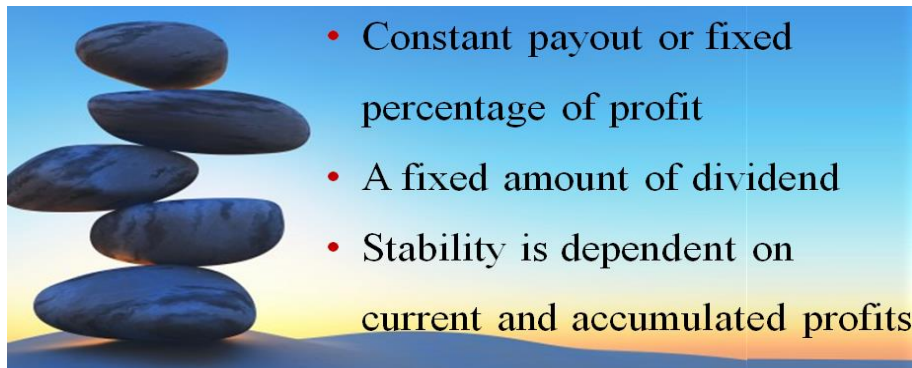
Dividend Policy, we decide on the basis that we have or have not investment opportunity. But in **Active Dividend Policy**, we make decision by considering many other things i.e., preference of shareholders, how much a company wants to grow, how many institutional restrictions on that business etc.

Module 13: Dividend Policy

Topic 234: Dividend Stability

In this lecture, we will be talking about the Stability of Dividend; what is meant by stable dividend, what is the significance of stable dividend and how important it is from the shareholders' perspective and also from the companies or businesses perspective.

Stable Dividend:



Constant payout or fixed percentage of profit:

So, let's first see that what is stable dividend. There can be two ways to decide about the Dividend Payout; either it can be a fixed amount that we pay every year irrespective of the fact that how much profit we have earned in that particular year and the other possibility is that we fix the percentage of profit and whatever we have earned, we pay that fixed percentage out as dividend.

For example, if we say that half of the profit will be paid every year i.e., if we earn Rs.10 per share profit then we will pay Rs.5 dividend and if we earn Rs.20 per

share profit then we will pay Rs.10 dividend. So, in this case, the dividend will keep on changing.

A fixed amount of dividend:

The second thing is that the **amount of dividend is fixed**. This means that **irrespective of the dividend a profit that we have earned, we will pay dividend as a fixed amount.**

Stability is dependent on the current and accumulated profits:

There are certain advantages or consequences of doing so but **dividend may be paid even if we do not have current profit because** the dividend is paid not from the current profits but **dividend is paid out of the retained earnings or the accumulated profits** or we can say that **Stability is dependent on the current and accumulated profits.** So, Stability of dividend will be more if instead of fixing the percentage of profit if we fix the amount of profit.

Payout:

Payout



- In stable dividend DPO is not constant
- Amount of Dividend is Constant
- In long-run DPO may be the same

In stable dividend DPO is not constant:

In stable dividend, **DPO (Dividend Payout) ratio is not constant because DPO (Dividend Payout) ratio is actually relates dividend per share with earning per share** and obviously **earning per share is not constant every year**. So, this means that **if earning will change then the amount of dividend** so, if the dividend is fixed or the **Dividend Payout ratio is fixed then we will not get a stable or constant dividend.**

Amount of Dividend is Constant:

The second thing is that to make the stable dividend we have to fluctuate the dividend payout but the amount of dividend will be constant. Suppose, if a company pays Rs.7 per share dividend every year and it remains Rs.7 every year for a longer period of time, we call it a Stable Dividend.

In long-run DPO may be the same:

If we are earning profit in certain years more and in other years a bit lesser and we fix the amount of dividend for a certain period of time then we may see that in a long-run (let's say a period of 10 years) when we calculate the average payout then we may come up with the same DPO which we thought that we should be giving and distributing to the shareholders.

Stable Dividend Preference:

Now, the Stable Dividend is Preferred as shareholders' preference is that dividend must be stable that they receive a certain amount regularly.

Stable Dividend Preference

- Shareholders prefer fixed dividend over time
 - Regular source of income
 - Certainty
 - Informational content (image of the company)
 - Institutional preference



Why shareholders want that their dividend should be fixed, get a fixed amount of dividend?

There are many obvious reasons for example:

Regular source of income:

Some people do investment for the purpose to get a regular income from that investment. If this income is in the form of dividend and the dividend is paid out regularly then it means that they will be getting monthly income regularly which they can use for their needs.

Certainty:

If a fixed amount of dividend is paid out every year then the element of risk is decreases. The return becomes more certain and people start relying on this return and that is a good thing for the company or organization.

Informational content (image for the company):

Similarly, a dividend which is stable or constant over a period of time also has an informational effect meaning that a thing spread in the stock market or economy that this company is financially strong and has the capability to pay a fixed amount of dividend regularly and it has an earning capacity to absorb the shocks of economy going up or down and can give a fixed monthly income to its shareholders.

That is an extremely important thing because that would increase the reputation of the company in the stock market and attract more investors and the share price is likely to go up.

Institutional Preference:

Institutional Preference also has to be considered as we know that there are two type of investors in the stock market; one is individual investors and others are institutional investors i.e., banks, mutual funds, financial institutions, large brokerage houses etc.

These are those companies or organizations which have more money and these organizations unlike individuals also have a set investment policy. That policy may be that the investment should be made preferentially in those companies which pay regular dividend then in that case, institutional investment will be attracted there.

We can conclude our discussion that a Stable Dividend is good for the company for the investor. To make stable dividend, we need to pay a constant or fixed amount of dividend every year and if we will do so then this will very important for the company.

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Topic 235: Why Dividends are Paid?

This is a continuous discussion on dividends and dividend policies and payments related to dividend.

Why Dividend? Why dividend should be paid? Why there should be any dividend?



There are many answers to that and there are many factors which influence that and we will be addressing these factors more and more in the next lectures.

To satisfy shareholders:

But the first thing is that in nutshell we can say that the dividends are paid to satisfy the shareholders. The shareholders are actually invest for return and return in the form of cash is the Dividend.

Regulatory compliance:

In many cases, dividend is to be paid for the Regulatory compliance. For example, we say that when a company is listed in Pakistan Stock Exchange then it has to follow some regulations and these regulations are from different perspectives

among one of them may be that this company pay dividend regularly and if it will not pay dividend on a certain period of time then this company may be de-listed from the regular stock exchange. So, these kinds of Regulatory Compliances may also be a reason why dividend should be paid.

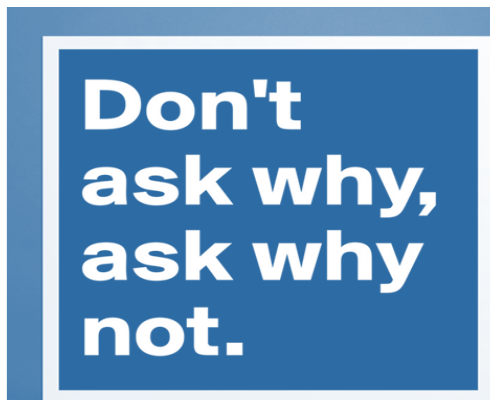
No Investment opportunity:

As we have talked about Dividend Residual that a company should pay dividend that time when if it does not have any opportunity to reinvest this profit further and this is how we say that when a company does not have any investment opportunity.

Why Dividends are not Paid? Or when dividends should be paid?

Why Dividends are not Paid?

- Investment opportunity
- Liquidity
- Control may be diluted if further shares are issued
- Binding covenants



Investment opportunity:

The first thing is that dividend should be paid that time when there is not investment opportunity. And its opposite is obvious that dividend does not pay that time when there is investment opportunity available. We have described Investment Opportunity that is; if a company has opportunity to earn more than the market rate of return then it is called **Investment Opportunity**.

Liquidity:

In order to pay dividend, a company should have Liquidity meaning that if cash has to be paid in the form of dividend then it is necessary to have cash and if that cash or liquidity is not present then it may be possible that this company may have

to borrow some money to avail that cash or to liquidate some assets and there may be consequences of that.

Control may be diluted if further shares are issued:

Similarly, if dividend has to be paid then the control may be diluted by paying that dividend if further shares are to be issued. It means that if a company has paid dividend through which a gap has created or it has deficiency of funds then it may be that this company has to issue further shares to cover that gap. And if further shares will be issued then there may be a possibility that there may occur difference in the holding of shareholders.

Binding covenants:

In some cases, when firms have taken some loans or financing from some institutions then there are certain requirements of the agreement that they have made which we call them the **Binding covenants** that is; the **agreement which buying a certain company to pay dividend only if certain conditions are fulfilled**.

It means that there may be one condition that if profit will be more than a certain amount then dividend will be paid or if the liquidity of the company will be at certain level then the dividend will be paid. So, that is also one reason why dividends at times are not paid.

We can conclude our discussion that **we have two possibilities** that why dividend should be paid for which we have some argument that we have to consider and similarly other possibility is that why dividend should not to be paid.

So, we have got both things in front of us and then we have to decide that considering both pros and cons or the arguments in the favor of dividend has to be paid and the restrictions why the dividend cannot be paid. So, we have to then decide by considering everything how much should be the amount of dividend in every year.

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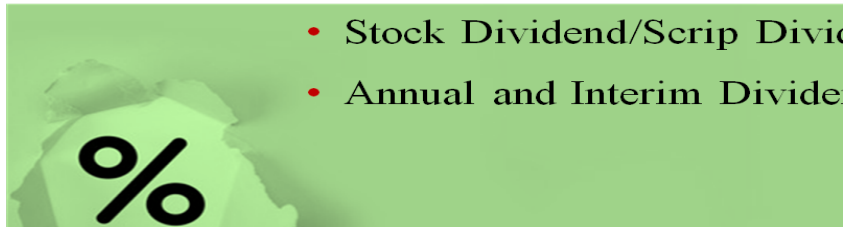
Topic 236: Types of Dividends

We are continuing our discussion on dividend and dividend policy and in this lecture, we will be talking about various types of Dividend.

Types of Dividend:

Dividend Types

- Cash Dividend
- Stock Dividend/Scrip Dividend
- Annual and Interim Dividend



Annual and Interim Dividends:

Some dividends are paid at the end of the year and the dividend which is paid at the end of the year is called **Annual Dividend** and the dividend which is paid during the year is called **Interim Dividend**.

But in addition to that, there are two other types of dividends which are very important;

Cash Dividend:

A type of dividend which is paid in the form of cash. Here it is important to remember that declaration of dividend and payment of dividend are two different events. A company first declared dividend and then after some time, the declared dividend is paid in the form of cash. So, this declared dividend can be paid in the form of cash therefore we call it **Cash Dividend**.

Stock Dividend/Scrip Dividend:

And this declared dividend can be paid to the shareholders in the form of the stocks of the company itself meaning that its further shares of such same value that the dividend has to paid, will be issued to the shareholders and we also name such shares as **Bonus Shares**. So, the **Bonus Shares or Scrip Dividend or Stock Dividend** is that dividend which is not paid in the form of cash.

There are some consequences of both dividend meaning that if there will be given Bonus shares then it will increase the number of outstanding shares and it also affect the shareholders' equity and book value. This effect of shareholder's equity and book value is also on the Cash Dividend because both the dividends (I.e., Scrip dividend or Cash dividend) are paid out of the retained earnings which is the part of equity.

Cash Dividend:

Cash Dividend

- Investors prefer cash
- Liquidity is required
- May lead to borrowing
- Reduces market price of the stock



Investor prefer cash:

The investors instead of taking further shares, they prefer to get cash that they may use for their needs.

Liquidity is required:

In order to pay Cash Dividend, we should have liquidity means we should have cash. Because it is not obvious that any company has any asset available in the form of cash so, liquidity is one concern when we have to pay Cash Dividend.

May lead to borrowing:

Likewise, if it has been paid Cash Dividend and we have to take further debt or to do borrowing and this is also an important point present in the decision of Cash Dividend.

Reduces Market price of the stock:

It is either Cash Dividend or Stock Dividend, then due to these dividends there has an initial dip in the market price but the announcement of Cash Dividend is a good news for the investors and when this announcement has happened then due to this announcement the share price also goes up in the stock market.

So, we can say that Cash Dividend decreases the share price but before it as a good news, the stock market goes up or increases prices of shares or stocks.

Another thing is that after some time of the payments of Stock Dividend or Cash Dividend, we have seen that market also again recovers the price of that share. So, Cash Dividend is considered to be an excellent thing.

Stock Dividend/Bonus Share:

Stock Dividend/Bonus Share

- Payment of company's shares as dividend
- Increases the issued capital
- Percentage holding does not change
- No cash outlay
- Market price goes down



Stock Dividend/Bonus Share is a good thing if we compare it not getting any dividend because our total value of investment increases. And if anyone wants cash then they can convert some of their shares into cash by selling them in the market

and for this in the discussion of dividend policy, we use the term **Homemade Dividend**.

How these Stock Dividends/Bonus Shares actually work?

Payment of company's shares as dividend:

The **Payment of company's shares is as dividend** and this is something which increases the issued capital meaning that when we give cash dividend then it is paid out of the retained earnings so it decreases equity and capital does not change.

Increases the issued capital:

But when we issue Bonus Shares then Bonus Shares are subtracted in the form of dividend from the retained earnings and then added in the share capital and usually in normal condition this does not affect our overall shareholders' equity but our two items of the equity (that is retained earnings and issued, subscribed and paid-up capital) change. It means that this balance is deducted from one account and added into the other account.

Percentage holding does not change:

And the **Percentage of holding does not change** because Bonus Shares are issued proportionally. It may be possible that if a company issued us Bonus shares and we have sold them in the market then our percentage holding will be affected.

No Cash outlays:

As this dividend is in the form of shares therefore there is no Cash Outlay.

Market price goes down:

The important thing that we have said in the Cash dividend also that market price goes down even with Scrip Dividend. Its logical reason is that our shareholders' equity which we divide with the number of shares and calculate the book value the there will be no changes in the value of equity from Bonus shares but number of outstanding shares get increased.

It means that now we after the Bonus Shares have to divide that equity with a more number of shares which decreases our book value. And the book value is actually the basis for the market price.

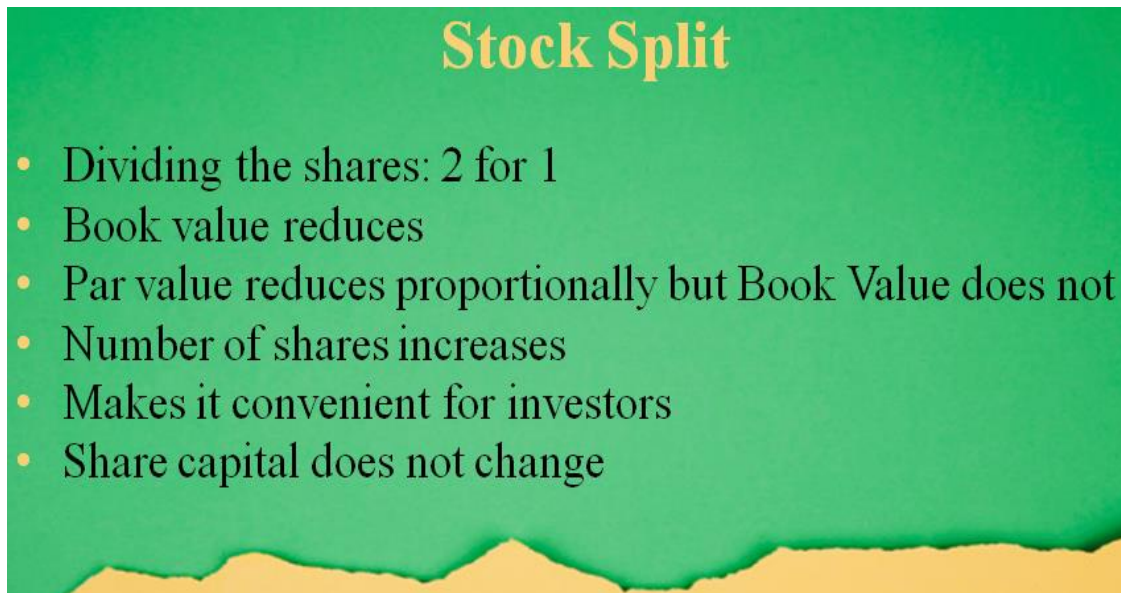
We can conclude our discussion that Dividends can be of two types; in the form of Cash known as **Cash Dividend** and in the form of bonus shares known as **Stock Dividend or Bonus Shares**. Both of them have their own requirements and pros & cons and there is some similarities and also some differences in between them. We have also seen that the preference of Cash Dividend is more but in the form of Bonus Shares it can be converted in the form of cash to some extent.

Module 13: Dividend Policy

Topic 237: Stock Split and Reverse Split

In this lecture, we will talk about Stock Split and Reverse Split and their details and effects.

Stock Split:



Stock Split

- Dividing the shares: 2 for 1
- Book value reduces
- Par value reduces proportionally but Book Value does not
- Number of shares increases
- Makes it convenient for investors
- Share capital does not change

Dividing the shares; 2 for 1

Stock Split means Dividing the share in more number of shares. This means that if we divide 1 share into 2 shares then this will be considered as a **Stock Split**. A share itself is also a portion or a small segment of the capital meaning that when we divide share capital into small parts then we get number of shares and if we further divide or break such parts or pieces of shares into more number of shares or pieces then it is called **Stock Split**.

So, if we are splitting 1 share into 2 that will be 2 for 1 is Stock Split and likewise we can split into more number of shares.

Book value reduces:

When Stock Split takes place, the book value also reduces. As we have seen that book value is actually our equity divided by total number of shares so, whenever the number of shares will get increased then it will definitely affect our book value and this will not change proportionally because there are other components in the equity.

Par value reduces proportionally but book value does not:

When we are splitting a stock into 2 or 3 or 4 then as a result certainly the par value also reduces. As we have discussed that **Par value** is such value of one share that if share capital is divided in a certain number of shares then the value of one share which we get is called the **Par value**.

It means that if we have share capital of Rs.100 and we want to divide it into 10 shares then the value of every share will be Rs.10 (i.e., $100/10 = 10$) which we call Par value. And if capital is Rs.200 and we want to divide it into 10 shares then the value of every share will be Rs.20 (i.e., $200/10 = 20$) which we call Par value.

So, it is very important that we should have the understanding that what is Par value, Book value and what is Market value. So, Par Value reduces but Book Value does not change with that proportion.

Number of share:

When the stock is split then obviously the number of outstanding shares increases.

Makes it convenient for investors:

The split is done for the purpose of making it convenient to the shareholders to invest in that company. Suppose if a share get expensive in the market lets it is of Rs.500 whose par value was Rs.10 then if we split that share into 2 shares then now the value of 1 share will become near to Rs.250 and people will be conveniently able to buy that stock from the market.

Share capital does not change:

The share capital amount actually does not change because the par value is reduced and the number of outstanding shares does change keeping the share capital constant.

Reverse Split:



Reverse Split

- Combining the shares: 1 for 2
- Par value increases
- Number of shares decreases
- High par value requires more investment
- Share capital does not change
- Book value is increased but not proportionally
- Reverse Split signals financial difficulties

Combing the shares; 1 for 2:

In addition to Stock Split, there can be a reverse process meaning that when we can divide stocks or shares into small pieces then we can also do reverse split process in any firm and that will be combing the shares meaning instead of dividend 1 share into 2 shares, we have to combine 2 shares and make it 1 share and that is called the **Reverse Split**.

If Stock Split means 2 for 1 then Reverse Split means 1 for 2 meaning that one who has 2 shares then these 2 shares will have to be taken and have given 1 share. And we can say as rule of Thumb that if it is such that then its value of 1 share will become the value of 2 shares.

Par value increases:

With Reverse Split, the Par value of the share increases and definitely it becomes twice if it is 1 for 2 shares (combining 2 shares and making them as 1 share) and if it is 1 for 4 then it becomes 4 times and likewise we can relate the ratio of the Reverse Split.

Number of shares decreases:

The number of outstanding shares will decrease because we have issued 1 share instead of 2 shares then if we have 100,000 outstanding shares then they will become 50,000 by decreasing and likewise we can relate the proportion.

High par value requires more investment:

As we have said that the reverse split of stocks is done for the reason that we have to make such investment possible for more people but if it is opposite of that the value of share is less enough and this is not attractive for the people then we can make their value more and can make them some attractive by combing these shares through **Reverse Split**.

Here another concept that we should keep in mind is that of **Penny Stocks**; if the market value of any share is less enough then it is called Penny Stock meaning a Stock of less money and there is more risk element in Penny Stocks.

Share capital does not change:

The share capital does not change even with Reverse Split because there has been changes occur in number of outstanding shares, in the par value but the total amount does not change.

Book value is increased but not proportionally:

The book value also changes but that increase book value may not be proportional to the stocks split whose reason is that there are other thing also in equity.

We can conclude our discussion that it is possible and companies do make such arrangements that they divided their shares into more than one share which is called **Stock Split** and at times companies also combines some the shares if the share price is very small and that is called **Reverse Split**.

Module 13: Dividend Policy

Topic 238: Stock Repurchase: Treasury Stock

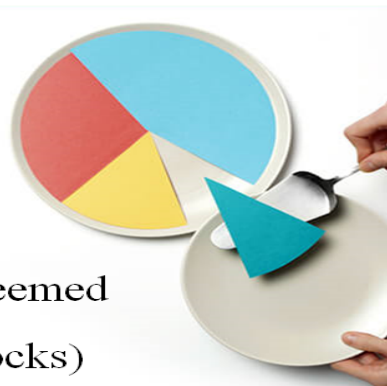
Treasury Stock is a very interesting concept and this is actually related to redemption of shares.

Stock Redemption:

Redemption means taking back. There is one possibility in some cases that securities issued by company may be taken back by the company or redeemed by the company.

Stock Redemption

- Common stocks cannot be redeemed
- Going Concern principle
- Bonds can be redeemed
- Preferred stock can be redeemed
(Redeemable Preferred Stocks)



Common stocks cannot be redeemed:

But in other cases, the company cannot do so and that is the case of Common stocks. Common stocks are issued for forever means that if a company exist and issue some common stocks then these common stocks will be issued till its existence and they cannot be cancelled by taken back or redeemed.

Going Concern Principle:

And this is because of the principle of Going Concern. The **Going Concern** means that the company will keep on functioning or operating in foreseeable future and there is no intention nor there is no any requirement of shutting down or closing the company. It means that if company will get running for forever then its common share will also remain outstanding.

Bonds can be redeemed:

On the other hand, debt securities like bonds can be redeemed.

Preferred stock can be redeemed (Redeemable Preferred stocks):

Likewise even Preferred stocks or shares can also be redeemed if these are issued as Redeemable Preferred stocks.

So, Preferred stocks and bonds can be redeemed but common stocks cannot be redeemed.

Treasury Stock:

The word **Treasury** means ‘Treasure’ or ‘Khazana’ but here Treasury Stock is used in quite different meanings.

Treasury Stock

- Purchase of company's own stocks
- Issued capital does not decrease
- Outstanding shares decrease
- Reduces the value of Shareholders' Equity



Purchase of company's own stocks:

When a company purchases its own shares from the market (Remember: this is not redemption or cancellation of shares) this is just repurchasing of the shares from the market.

Issued capital does not decrease:

But these shares remain as the outstanding shares of the company in terms of the issued shares. It means that free float or outstanding shares available for trade in the market must be decreased but issued shares do not decrease.

Outstanding shares decrease:

The outstanding share capital meaning those numbers of shares which can now be traded in the stock exchange get decreased because that this company has purchased some shares and kept with itself.

Reduces the value of shareholders' equity:

All this is reduces the shareholders' equity because the amount of Treasury Stock is subtracted from the equity and we add retained earnings in issued capital then we get shareholders' equity. So, when **Treasury Stock** is subtracted from our equity then our book value and shareholders' equity get decreases.

Why Treasury Stock or Why shares are purchased and kept with itself?

Why Treasury Stock?

- Increases the return of the remaining shareholders
- Distribution of funds when there are not attractive investment options
- May have a positive signaling effect
- Affects the percentage interest in the company



Increases the return of the remaining shareholders:

Treasury Stock increases the return of the remaining shareholders like if there were 1,000 outstanding shares and the company has taken 200 as Treasury Stocks so, lesser number of outstanding shares are there with the people and the next year profit will be distributed in those lesser number of shareholders and hence the profit will be enhanced.

Distribution of funds when there are not attractive investment options:

It means the such funds or money or profit which we have earned and for this if there is no good reinvestment opportunity then we can purchase our own shares and can be increase the benefit of other shareholders.

May have a positive signaling effect:

This may also have a positive signaling effect meaning that to create or take Treasury Stock gives such signal in the market that a company has available profit or cash and company wants to give more benefit to its shareholders.

So, collectively we can say that **Treasury Stocks** are our own shares which have been purchased from the market and then that company holds these shares with itself. We can also say that the presence of Treasury Stock is not bad in the equity of any company and this may be a positive thing and a positive signal is also generated out of that.

Affects the percentage interest in the company:

The last thing about the Treasury Stock is that if a company has taken or purchase some Treasury Stocks then it does not mean that it cannot be resold in the market and whenever company need financing then these shares can be again resold on the market price and in that case Treasury Stocks will be eliminated out of the equity of that company.

Module 13: Dividend Policy

Topic 239: Retentions or Plough Backs

We have already discussion on Retention and Retention Ratio, Dividend Payout Ratio and in this lecture; we will talk about some other aspects of Retention.

Retention:

Retention

- Portion of profit not distributed as dividend
- Profit is an internal source of financing
- Return of shareholders should be maximized
 - By distributing
 - By retaining
- Investment Opportunity



Portion of profit not distributed as dividend:

The profit which is retained by the company is for the purpose of reinvestment in the business and that amount of profit which is not distributed among the shareholders is called **Retained Profit or the Retention** of the company. As more profit will be retained, more will be the chances of growth of company.

Profit is an internal source of financing:

This **Profit** is an internal source of financing whereas external source of financing is **equity or debt** i.e., to take money from shareholders or from bond holders for the company but when company uses that money taken as debt and reinvest it in its business and increases its capital then it is called Profit and it becomes an internal source of financing for the company.

So, it is a source of financing, it can be or should be used for the investment purposes. Once this profit is reinvested then Retention maximizes the growth.

Return of shareholders should be maximized:

The return of the shareholder is to be maximize and this is for sure as we know that the whole financial management actually focuses on one thing that is maximization of return. And every financial decision should be made by keeping in mind that the returns are to be maximized.

Will it be maximized By Distributing profit or it will be maximized By Retaining the profit?

If we want to maximize the return and the growth, we have to retain provided we have got an investment opportunity.

Investment Opportunity:

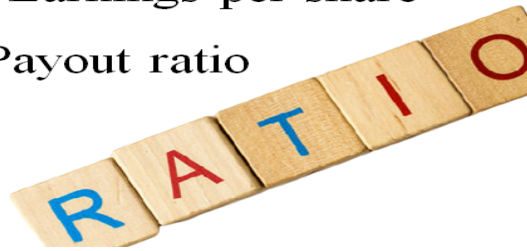
We have discussed about it earlier that if a business has an opportunity to earn more return as compare to the market then this is called **Investment Opportunity**.

Retention Ratio:

Retention Ratio

- Profit retained to profit earned
- Retention per share to Earnings per share
- One minus Dividend Payout ratio

$$RR = 1 - DPO$$



Retention is measured in terms of Retention Ratio. The Retention Ratio is actually a ratio of such profit which has been retained with that profit which is earned meaning that it is the ratio of the profit retained and the profit earned or the retained profit divided by the earning per share.

If we have got Retention ratio then it will be Retained profit or Retention per share over Earning per share.

And if we have already calculated the amount of dividend and with that we have calculated the Dividend Payout Ratio (I.e., **DPS/EPS**) then Retention Ratio can also be calculated by subtracting the DPO (Dividend Payout Ratio) out of 1.

Formula to calculate RR (Retention Ratio) when DPO (Dividend Payout) ratio is given:

$$RR \text{ (Retention Ratio)} = 1 - DPO$$

We can conclude our discussion that **Retention** is important as it related with the growth of any company. In order to calculate the **Retention Ratio** either we have to relate amount of retention with the retained earnings or if we have already calculated the DPO (Dividend Payout) ratio then we can also calculate Retention ratio by subtracting this amount from 100% or 1.

More will be the Retention Ratio, more will be our expectation about the growth and less will be the Retention Ratio, less will our expectation about the growth of

any company. And if Retention Ratio get extreme or becomes 100% then we will expect maximum growth or if Retention Ratio becomes 0 (zero) then the growth of any company will also become zero (0) and that is the importance of retaining profit and calculating Retention Ratio.

Module 13: Dividend Policy

Topic 240: Investment Opportunity

Investment is a very complicated process and when we use the term **Investment** then generally we are referring to the shareholders or the bond holders. But once a company gets money or funds from these shareholders or the bond holders or some other external sources then the company also has to invest that money in some operations in order to generate profit.

There is a profit expectation or an expectation of return of the fund providers that is the shareholders and the bond holders.

Why investment should be made?

Why Invest?

- To earn profit
- Preference is to get maximum profit
- Company should make prudent decisions for maximum return to shareholders



To earn profit:

Investment is made to earn profit to enhance the return.

Preference is to get maximum profit:

This profit again we need to maximize this profit. Every decision of investment should be made keeping in view the point that is to be maximized.

And every such investment when we do comparison as we have discussed in Capital Budgeting where we have talked about that if we have more than one investment opportunities then how we will have to set preference in order to maximize our profit or return or wealth.

Company should make prudent decisions for maximum return to shareholders:

So, while investing, the management or the decision makers need to make the most prudent decision moves which will maximize the return of the shareholders.

Investment Opportunity:

We need to have Investment Opportunity and we have again touched upon this topic that what is Investment Opportunity.

Every business has an investment opportunity and that is why the business is started. We can say that every company or business when it is registered or started they have an investment opportunity but technically when we use the term ‘Investment Opportunity’ means that the company should have an opportunity to put that investor’s money in an activity in a business where they will be available to maximize their returns.



It means that how much return or profit can be given to the shareholders if their profit or return have to be given or distributed in the form of dividend. If the shareholders have received money and we suppose that these shareholders instead of consuming that money, again invest that money somewhere then how much return they may get from that investment which we call generally market rate of return.

And in contrast to that, whenever company will invest this amount somewhere itself then this company may earn how much rate of return on this amount and has capability to give this return to the shareholders. So, this is the decision and this is the concept of Investment Opportunity.

Excess Return

- Market Rate = 14%
- Return from reinvestment in the Business = 19%
- Profit may be reinvested

Suppose the market rate prevailing is 14% and the company has an opportunity to invest in its own business at 19%. So, there is an investment opportunity which means that the shareholders can earn 14% if money is distributed among them. And if this money has to be retained then company will earn 19% for the shareholders and given them and that is **Excess Return**.

In this case, we may say that the profit is to be reinvested.

It is very important to keep in mind that an investment opportunity and excess return is always with reference to the general opportunities that are available to the shareholders.

Module 13: Dividend Policy

Topic 241: Considerations for Dividend

In this lecture, we will talk about that why dividend has to be paid or what do we consider when deciding about the payment of dividends. The management has certain considerations and some of these we have already discussed earlier so we will emphasize more on those considerations we have not touched upon.

Dividend Considerations:

Dividend Considerations

- Investors' preference
- Investment opportunity
- Availability of profit
- Taxes (Double taxation)
- Financing cost
- Regulatory requirements
- Liquidity



Investors' preference:

Dividend and particularly Cash Dividend is the preference of the shareholders whose reason is that there may be some dip or decrease in price after adding dividend in the share price but the increase before adding dividend in share price and the recovery in the share price after some time has to pay dividend as an extra return effectively paid to a shareholder.

Investment Opportunity:

The second that we have to consider is Investment Opportunity which we have discussed earlier.

Availability of profit:

We have discussed that when we are considering the payment of dividend we need not to have profit in that particular year.

It means that if a company has not earned profit in any year then it has to pay dividend from its accumulated profit or retained profit then it is extremely positive sign for the company but there has to be profit meaning that one of both profits (i.e., current profit or accumulated or retained profit) should always be available for the distribution of dividend and if none of these will be available then dividend cannot be paid.

Taxes (Double Taxation):

Another important element is the Taxes whenever a company earns profit then we know that we firstly calculate the Profit before tax and then we deduct the amount of income tax as per the rate applicable to that company and we get the figure of Profit after Tax.

Now, after calculating this profit or final amount of profit (PAT or Profit after tax), we distribute a part of that as dividend among shareholders among but in many cases these shareholders when they get dividend then they also have to pay tax on it. And this is the concept of **Double Taxation.**

That is once the company has paid the tax on the profit then the distribution of profit should not involve any additional taxes but yes, this is a consideration so, if dividend is paid and there will be double taxation so their return will be reduced.

Financing Cost:

While paying dividends, the management should also consider the Financing cost because if liquidity is not there and the funds are to be borrowed then in that case financial cost will become pertinent and relevant thing in the decision making. So,

if we have no cash and we have to borrow money then the interest that we have to pay on that amount will affect the profit of our next year. So, that is also one consideration for the payment of dividend.

Regulatory requirements:

Then there are regulatory requirements meaning that regulatory requirements in which dividend has to be paid strictly or it become compulsory for us to pay dividend. The rules or regulations which are made for stock market should also need to be considered during the payment of dividend.

Liquidity:

And finally we say that there has to be liquidity if cash dividend is to be paid and this part is only related to cash dividend. It means that if there is not liquidity then dividend can also be paid but that will be in the form of bonus shares or scrip dividend.

We can conclude our discussion that it is necessary to consider many things while paying dividend; presence or absence of investment opportunity, profit, tax rates applicable, individual or corporate or company taxes, what will be the financing cost if borrowing money from somewhere, if there are regulatory requirements due to which it is necessary to pay dividend then a company or business has to decide to give dividend by considering all these things.

Module 13: Dividend Policy

Topic 242: Relevance of Dividend

In previous lecture, we have talked about some considerations of dividend that which things we need to consider during the payment of dividend.

Now in this lecture, we will discuss that the payment of dividend is how much relevant from the management, decision maker and especially from the shareholders' point of view.

Relevance of Dividend:

Relevance of Dividend

- Preference of cash dividend (Bird in hand concept)
- Regular income
- Tax consideration
- Financial signaling (market price)
- Home-made dividend is not practical



So, there is a theory or approach or school of thought which says that **it is Relevant to pay dividend** and we cannot consider dividend payment as irrelevant decision **whose reasons are given below:**

Preference of cash dividend (Bird in hand concept):

The Preference of cash dividend is also supported by ‘Bird in hand concept’ which is such a phrase about which we say that Bird in hand is worth two in the bush meaning that the thing that we have in our hand or we get today is more better than the thing which we will get tomorrow or have hope to get it tomorrow.

So, whatever we get today is preferred over what we may get tomorrow and that is why there is preference of Cash Dividend.

Regular income:

Cash dividend is relevant because it also given regular income. So, if we need cash for our expenses then we need to have dividend and if we will not get cash dividend then we cannot fulfill our expenditures due to increase in our share price and so, as dividend is a regular income for many investors therefore there is relevance of dividend.

Tax consideration:

The presence or absence of tax also make dividend as relevant meaning that if taxes are imposed in such a way that if the payment of dividend is harmful for

investor then it should not be paid and if the payment of dividend is beneficial for the investors then it should be paid. So, in any case, if there are income taxes, individual and corporate taxes, this tax is an important and relevant element.

Financial signaling (market price):

The payment of dividend is also a financial signal meaning that the dividend paying company is considered to be a good company who is financially strong, there are less chances of its bankruptcy and pay regular dividend and also have less element of risk.

It means that if a company wants to be a Blue Chip company (where Blue Chip means that the best companies in the stock market) then payment of dividend is one aspect which gives that signal and hence it becomes a very relevant decision.

Home-made dividend is not practical:

Home-made dividend concept is that if the company does not pay dividend suppose our share price was Rs.100 and the company earns profit of Rs.10 but it does not distributed as dividend then our share price will go up from Rs.100 to Rs.110 meaning that the increase which we have to get in the form of cash but we have received in the form of that increase in our share price. When our share was of Rs.100 and if we get dividend of Rs.10 then we will have total Rs.110 and now we have no cash but our share's value is now Rs.110.

And if we sell that additional portion in the market then we can convert it into cash as well and this is called **Home-made Dividend.** But there are several observations or question on this Home-made dividend which actually says that Home-made Dividend concept is not as simple.

We can conclude our discussion that there are several things which make dividend a Relevant decision and these are preference of the shareholders, the Bird in hand theory, source of regular income, the consideration of taxes and the signal of financial strength of the company and also that the Home-made Dividend is not a practical or a realistic thing.

Module 13: Dividend Policy

Topic 243: Theory of Relevance of Dividend-1

We are going through the discussion of Relevance of Dividend.

Why dividend is Relevant?

Theory of Relevance of Dividend:

The first or the basic thing is that it affects the value of a firm so, if the market value of a firm is affected by the payment or non-payment or increase or decrease of dividend then it becomes Relevant.

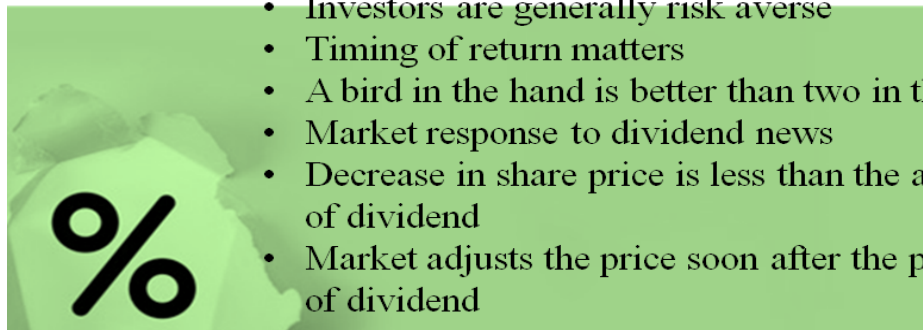
Or

The Relevance theory of dividend argues that dividend decision affects the market value of the firm and therefore, dividend matters.

But there are many other reasons why dividends are considered Relevant and this Theory of Relevance of Dividend or the Theories of Relevance of Dividend talk about that.

Theory of Relevance of Dividend

- Dividend decision affects the market value of the firm
- Investors are generally risk averse
- Timing of return matters
- A bird in the hand is better than two in the bush
- Market response to dividend news
- Decrease in share price is less than the amount of dividend
- Market adjusts the price soon after the payment of dividend



- Dividend decision affects the market value of the firm and this is the prime or most important part of it.

- But Investors are generally risk averse or they do not like to take risk. And as a result they would prefer whatever is more certain and that is today's dividend.
- Timing of return also matters meaning that if we have given 10% return today and we also will get 10% after one year from now then we would prefer to get that return today as compare to a year from now.
- A bird in the hand is better than two in the bush which means that whatever we have today in our hand is certain and it is preferred over what we are likely to get in the future.
- Another very important aspect is that Stock Market response of the investors to the dividend news that means when dividend is announced or its news comes then the price goes up in the market which is certainly a benefit for the investors or shareholders.
- As dividend is paid because dividend reduces the book value of a share, the market price originally also falls but the decrease in share price or market price is usually less than the amount of dividend. And hence a shareholder or investor gets benefit out of it which makes it a more relevant thing.
- Market adjusts the price soon after the payment of dividend that the market price goes back to the original position where it was before the payment of dividend meaning that whenever the dividend will be paid then market price will go down but after some days as market by recovering will get its original position and the amount of dividend becomes an extra return which makes it very relevant.

Walter's Approach:

Walter's Approach

- If ROI is greater than ROE then retain earnings
- If ROI is less than ROE then do not retain earnings
- If the two rates are the same then the company should be indifferent



Walter actually presented an approach regarding dividends which is a comparison of the earning power of the company as compare to the Return on Equity (ROE) or what the shareholders can earn.

Walter simply said that:

If the ROI (Return on Investment) or earning power is more or greater than ROE (Return on Equity) then the company should retain that profit or earnings and should not pay dividend.

If the ROI (Return on Investment) or earning power is less than ROE (Return on Equity) then do not retain earnings meaning that the money should be given to shareholders through which they can earn more profit or benefit.

If the two rates (ROI and ROE) are the same then the company or decision makers should be indifferent about the payment or nonpayment of dividend then it will not affect the shareholders that they get dividend or not.

Walter's Formula for the Value of Equity

Walter also has given a formula of the value of equity and this is the formula that we have already discussed while discussion the methods of evaluation and cost of capital as well.

$$VE = D / (KE - g)$$

Where, VE = market value of equity shares

D = initial dividend

KE = costs of equity and

g = expected growth rate of earnings

We can conclude our discussion that once we look at the relevance or irrelevance of dividends then there are several points in favor of both of them but dividend is certainly relevant in the way that there are many factors which affect the dividend or its relevance. And Walter also gave a formula and also an approach towards the payment or nonpayment of dividend which is also a part of dividends policy.

Module 13: Dividend Policy

Topic 244: Theory of Relevance of Dividend-2

In the first part of the Theory of Relevance of Dividend, we discussed that there are many reasons why we say that the Relevance of Dividend exist and we have also seen its Walter formula and here we will talk about another approach given by Gordon

Gordon's Approach:

- The Gordon says that basically the shareholders are risk-averse where risk-averse means they do not like to take risk.
- And that is why, they prefer cash today which is certain and not risky as compare to the cash or profit or capital gain or return which they can get in the coming days or future that is expected return. So, this certainty because of being risk-averse, this is what they prefer.
Or we can say that Today's cash is better than future capital gains, thus a "bird in the hand is worth more than two in the bush".
- Gordon says that investors being more of the nature that they prefer today's return and the risk element which is uncertainty so that is why, they go with the present return and that is the current or present cash. Or it resolves investor uncertainty (future is uncertain).
- Other investors prefer even lesser amount or income today which is more certain and not risky as compare to a bigger amount or income tomorrow or in the future.
- Similarly, Gordon says that due to the less risky nature of dividend, investors will discount the firm's dividend stream at a lower rate of return, 'r', thus increasing the value of the firm's share and hence the dividend becomes relevant.

Assumptions of Gordon's Model:

The Gordon Approach is based on certain assumptions that we need to keep in mind that are given below:

- All equity base firms mean that there is no leverage or debt.
- There is no external financing also and it is only the RE (retained earnings), the profits which are kept in the business by the company.
- Internal rate of return of the firm remains constant which means that the efficiencies of the firm and cost effectiveness are the same.
- Cost of capital remains the same regardless of the risk complexion and because we say that there is no external financing and external factors are constant so, definitely the risk complexion will also not change.
- There is another basic assumption of Gordon's model which we have seen other models also is Going concern that is; the firm will be Going Concern meaning that there are no chances of shutting down or closing or there is no intention to close this firm and in accounting we call it Going Concern.
- The retention ratio is also constant that means if 50% dividend is paid and 50% is retained then this ratio is also constant.
- There is another assumption of Gordon's model which is not a realistic one that the Corporate tax does not exist.

Gordon's Model: Growth Rate

Gordon gave this model of the Growth Rate of a firm and says that retention is something that determines the growth rate as we know that Retention is opposite of Dividend Payout (DPO) i.e., if 60% of the profit is distributed as dividend then remaining 40% will be our retained profit or retention. So, more will be our retention ratio, more will be the growth expected of the firm.

In other words, we can say that Dividend Payout (DPO) ratio will affect the growth of the company and ultimately the value of the firm.

Thus the growth rate (g) is also constant:

- Growth rate = retention ratio x cost of capital
- $g = b \times r$
- $g = 40\% \times 15\% = 0.4 \times 0.15 = 0.06 = 0.06 \times 100\% = 6\%$

We can conclude our discussion that we can say that Gordon actually presented a formula or a method or an approach which says that the dividends are relevant and the relevance is with reference to the growth of the firm and this growth is dependent on how much is paid as dividend and how much is retained. And more is retained by the firm and more is likely to be the growth of the firm.

Module 13: Dividend Policy

Topic 245: Theory of Relevance of Dividend-3

In this lecture, we will look at Theory of Relevance of Dividend from another couple of different approaches; something that will again help us to look at the dividends as something which are relevant.

Dividend Capitalization Model:

- The first one is Dividend Capitalization Model and this model states that the market value of a share or a company is actually equal to the present value of the future Cashflow streams in the form of dividends. It means that if we discount or find present value of those dividends which we have to get in the future then it is the market value of that share or company or stock.
- So, if the value of the firm is dependent on the future dividends then certainly the dividends are relevant.

Formula:

The formula to which we calculate the market value or the price of a share or a firm and this can be calculated by both ways meaning that the price of one stock can be calculated and to determine the value of overall firm.

$$P = E (1 - b) / KE - br$$

Where:

- P = Price of a share

- E = Earnings per share
- b = Retention ratio
- $1 - b$ = Dividend payout ratio (as ' b ' is in percentage therefore when we subtract it from 1 means 100% then we will get DPO)
- KE = Cost of capital or the capitalization rate
- br = Growth rate (here ' b ' is retention ratio and ' r ' is cost of capital)

Signaling Theory:

Another theory or approach for the Relevance of Dividend is the Signaling Theory. There are many factors, actions manifest or signal something means they convey a specific message to the people.

- The Payment of dividend is such thing through which potential investors (those people who can invest in that company) and those people who have already invested or purchased shares sends a message. So, Payment of dividend sends signals of strength and future growth of the company so, certainly dividends are relevant meaning that through this signal, if our investors or shareholders will consider our company strong and growing then market value or price of our company will also go up in the future.
- If any company or firm make changes in dividend policy (it is in its basics is that how much percent profit will be distributed as dividend and how much percent profit will be retained) or we can say that change in the dividend policy sends a signal:

Positive Signal: If more amount of profit is paid as dividend that sends a positive signal and this is how the investors perceive or the market price of a company goes up.

Negative Signal: Similarly, signaling theory states that there can be negative signal meaning that if any company does not pay dividend even if it is earning more profit then it generates negative signal in the market and people consider that this company is not financially strong or having liquidity issues or due to some other issues this company is not able to pay dividend.

We can conclude our discussion that this lecture is also about the **Relevance of Dividend** and **we have discussed here two things**; one thing is that **we find the value of firm by discounting the future dividends therefore dividend is relevant**. Other thing is that **payment or nonpayment of dividends and more or less payment of dividend sends signals**. The **payment of more consistent dividends sends positive signals** which show that this company is financially strong, stable and growing.

Module 13: Dividend Policy

Topic 246: Bird in Hand Theory

This Bird in Hand Theory is actually another clarification or another argument with reference to the dividend policy.

We have talked about the Relevance of dividend that why dividend is relevant about which we have talked about many approaches or arguments.

Bird in Hand Theory:

A very strong argument that we have discussed in previous lecture which we name “Bird in hand theory”

This phrase is very old that ‘a bird in hand is better than the two in the bush’ meaning that the **thing which we have in our hand today even less is more better than the things** which we will have **in the future** or which is difficult to get or whose chances are no or less to get.

What about Bird in Hand Theory?

- This theory of Bird in hand is applied to the dividend policy with reference to the fact that **today’s income or today’s return is better than that of tomorrow.**
- It is said that Dividend is an estimated value or **Dividends make up about 40% of the total stock market returns** earned by the investor is of

dividend. So, that means dividend is a very large proportion of returns although it is considered that most of the profits are made by capital gain but dividends do contribute a lot.

- **A bird in hand is better than two in the bush: Lesser amount in hand today is better than more with the company.** It means that when this dividend is considered as better because the future return or future capital gain is earned certain so, we apply this theory of Bird in hand in a way that the dividend we get today from the company in the form of cash is better than this amount of profit retained by the company and we get this appreciation of our stock or dividend in future. It means that if a company does not pay dividend then as a result the value of that company's share get increases. And this is that amount which a shareholder will get more in the form of market price in the future but this market price will be more or less in the future because of some factors and this is something which is uncertain and that refers to the 'Two Birds in the Bush'.

Criticism:

There is some criticism on this theory 'Bird in hand theory' meaning that this theory tells us about the Relevance of Dividend because 'Bird in hand is better than the two in the bush' so, dividend of today is better than the capital gain of tomorrow.

- But the first criticism on 'Bird in hand' theory is that **reinvesting dividend (they invest the money they get)** or what do investors do with the dividend they get that it is seen that they generally reinvest that money and the reinvesting the money means that they actually did not require cash for their internal purposes so, if they have to reinvest again in the business then why to pay dividend to them.
- Second criticism is about the implication of tax or **tax minimization (tax on dividend today reduces capital).** It means that when dividend has been paid then we have to pay tax on it and this tax is deducted from the dividend which we have to get today and hence the capital of the investor is reduced.

- Third one is **by asking for dividend they do not act rationally**. It means that because they have to reinvest their dividend in the company or any other company because it reduces their capital so, if the investors ask for dividend, if the investors say that they are happy with a lesser amount of dividend or return than more return in the future then they are not acting rationally. Here **rationally** means that they are not making investment with the thinking of maximization of their profit and this is not considered as a positive or favorable point.

We can conclude our discussion that Bird in hand theory is another theory in favor of Relevance of Dividend which states that the dividend getting today is certain even less therefore is better than that future increase in the value of share which may or may not be there. But there are **three basic criticisms**; investors get dividend and reinvest this dividend then why we need to pay them dividend, by paying dividend they have to pay more tax today which is not good and investors would not be acting rationally if they prefer lesser return today as compare to the more in the future.

Module 13: Dividend Policy

Topic 247: Dividend a Passive Residual

We are continuing our discussion on Relevance and Irrelevance of Dividend and also on the Dividend policy as a whole.

The Residual Theory of Dividends:

- Dividend paid by a firm is viewed as a residual
- The amount remaining or leftover after all acceptable investment opportunities

Another approach towards Dividend Policy is that dividend should be the Passive Residual where **Residual** means the leftover profit or remaining profit which should be paid among shareholders.

It is said that every business or firm has investment opportunities meaning that one opportunity where we can earn a lot of return for the shareholders by investing our profit there.

So, if a firm pays dividend then it should only be after what is left by investing our profit in that particular opportunity and that is the Residual and that is something which should be paid as dividend and this is what the Theory says.

Stages of Dividend:

This Residual Theory of Dividend actually step by step takes us to the point where we can determine the exact amount of dividend that can be distributed among the shareholders.

Stages of Dividend:

1. First determine optimal level of capital expenditures
2. Estimate equity needed to support the expenditure
3. Reinvested profits is utilized to meet the equity requirement
4. If there is a surplus that is, the residual, it is paid as dividends



1. First determine optimal level of capital expenditure:

Firstly we need to determine that what is the optimal level of capital expenditure meaning that what is such best combination of capital expenditure for the company in which it will get more benefit that we call Optimal level of Capital Expenditure.

2. Estimate equity needed to support the expenditure:

Once we determine that Optimal level of Capital Expenditure then we will see that how much equity will be required for that project. It means that when we have identified a project then second step is that because every project has a combination of debt and equity then we will see that what amount of equity will be required and that equity will be generated from the profits which are earned by the company.

3. Reinvested profits are utilized to meet the equity requirements:

Then the reinvested profits are utilized to meet that equity requirement.

4. If there is a surplus that is, the residual, it is paid as dividend:

Then if there is surplus meaning that after doing optimal investment and fulfilling the requirements of equity and if we are left some profit or amount with us then that leftover amount can be distributed as dividend among the shareholders. This is what the Residual theory says that.

The Residual Theory of Dividend Formula:

Thus, the formula of dividend in Residual Theory is:

Dividend amount = Net income – Additional equity needed

Example

- A project required Rs. 120 million. Equity ratio is 70% and profit is Rs. 95 million. How much Dividend can be paid?

Solution:

Additional Equity = $120 \times 70\% = 120 \times 0.7 = 84 \text{ M}$

Dividend = Net income – Additional equity needed

Dividend = $95 - 84 = 11 \text{ M}$

We can conclude our discussion that it is said in **Residual Theory of Dividend** that amount of profit should be given as dividend to the shareholders which that company cannot reinvest this amount of profit in its own project meaning that for which amount, company does not have investment opportunity.

And then **investment opportunity** is then described or defined as the opportunity of reinvesting the profits in the firm or company at a rate higher than that can be earned by the investors themselves.

So, this **Residual Theory** actually states that dividend should only be paid if there is surplus or residual cash or profit available. And if there is no cash or surplus available even for 10 or 20 or 30 years then company should not have to give dividend and it is totally different from the approach that we have seen in the Relevance of Dividend.

Module 13: Dividend Policy

Topic 248: Theory of Irrelevance of Dividend-1

In our previous lectures, we have seen that many people have given many arguments about the Relevance of Dividend (I.e., Bird in hand theory, Residual Dividend policy etc.) and they are considered also.

The Irrelevance Theory:

Now we are going to talk about the Irrelevance of Dividend meaning that the theory or arguments which says that the payment or nonpayment of dividend is indifferent.

There are two things which are more important which we should keep in our mind that; one is that what is the advantage or disadvantage of payment or nonpayment of dividend to the shareholders and other is that what is impact of that on the firm or company. The **Irrelevance Theory** says that none of the two are affected by this.

- The first thing which is categorically said is that dividends are not relevant.
- The argument is that a company's ability to earn and grow determines the market value and drives the stock price; not the dividend payments.

It means that how much that company can earn and how much it can grow is something that determine the market value; how much this company retain and how much has been distributed as dividend among the shareholders from that earned profit is not something which affects the value of the firm. So, the firm should concentrate more on earnings and growing if it wants to increase their value.

This Irrelevance Theory has its own arguments which are given below:

- The first argument is that there is no impact of payment of dividend on the stock price.

- **Dividend does not affect a company's growth.** As we have seen in Relevance Theory that the growth is a function of retention means that more amount of profit is retained, more is likely to be the growth of the company. But the Irrelevance Theory of Dividend is against that Relevance theory.
- **Cash dividend is offset by external financing.** It means that if a company has paid cash in the form of dividend then the gap created from that dividend payment can cover by external financing or external equity and there will be no difference to the company that it has retained its own profit or it has taken external financing because both incur cost meaning there is cost of equity on retained profit and if we have to take external financing then there will also be cost incurred.
- **The wealth of an investor or value of total investment is not affected by the payment or non-payment of dividend.** If we have paid dividend then market price of our stock, first the dividend is our total wealth and if dividend is not paid then the current market price is the total wealth of the investor.
- So, this Irrelevance Theory concludes that the split of Dividend and Retained Earnings has no impact.

We can conclude our discussion about **Irrelevance Theory** which states that as the payment of dividend does not affect the wealth or share price of the investor and also does not affect the growth or profits of the company therefore **Dividend is not relevant or irrelevant.**

Module 13: Dividend Policy

Topic 249: Theory of Irrelevance of Dividend-2

We have started our discussion on Dividend Policy in the previous lectures and we have discussion arguments in favor of Relevance and also some arguments in favor of Irrelevance of Dividend. In this lecture, we will also talk about some of the arguments in favor of the Theory of Irrelevance of Dividend.

The concept and the effect of this concept, we will see in this lecture is Dilution.

Dilution Defined:

The word ‘Dilution’ if we use commonly then it means to reduce the efficacy or strength of anything i.e., if we have got a solution or mixture and we add water in it then we say that the quantity is increased but the efficacy or the strength has reduced.

When we use this concept of **Dilution in finance** then it means that when a company issues more shares, as a result the book value of the share decreases. It means that when in our shareholder’s equity, the number of shares have been increased then our share capital increases but there has not been increases in our retained earnings or reserves etc. due to which our book value per share decreases and we call this concept ‘**Dilution**’.

As a result of Dilution, the claim on the assets of the company per share reduces and later on when the company earns income or profits that income or profit per share also reduces. So, **Dilution** is actually reduces the shareholder’s claim on asset and their per share earnings.

Or

The decrease in the proportional claim on earnings and assets of a share of common stock because of the issuance of additional shares.

Dilution does not affect:

Now, the **argument in the Theory of Irrelevance** is that actually nothing happens with Dilution. It means that when we issue further shares because of the reason that we have paid dividend which will create gap meaning it will need further financing and that further financing will be generated by issuing more additional shares.

Now, as a result **Dilution will take place**.

The **Theory of Irrelevance** says that once the Dilution will take place and book value will go down and as a result the market value of the company's share will also go down but this will not affect because if we add the value of dividend in that reduced value of share then that value collectively will become equal to the same value that this company had before the addition of dividend value and before additional financing.

Or

The sum of discounted value per share of common stock after financing **plus** current dividends paid is exactly equal to the market value per share of the common stock before the payment of current dividends.

So, as a result we can say that Dilution does not affect.

Result of Dilution:

The Irrelevance Theory says that whether there is dilution or there is no dilution or there is payment or nonpayment of dividend, the shareholders will remain indifferent.

Or

Shareholders are indifferent between receiving dividend and having earnings retained by the firm.

Why the shareholders will remain indifferent in case of dilution or payment/nonpayment of dividend?

Homemade Dividend:

It is because of the fact that they can make a Homemade Dividend. This term of Homemade Dividend is very interesting in which we see two things;

- If shareholders will be given less dividend than their requirement then they will sell some part from their shareholding and will create dividend for themselves.
Or if dividends are lower sell some shares to get desired cash distributions.
- If shareholders will be given more dividend than their requirement then they will reinvest that additional dividend which is more than their requirements and in this way, the shareholders are in the position to adjust the amount of dividend by their selves.
Or if the dividends are higher, purchase additional shares of stock in the company.
This is the argument of Irrelevance Theory.

We can conclude our discussion that where **Irrelevance Theory** gives us many arguments there is an argument is Dilution and Dilution is actually the issuance of further or additional shares. The need for issuing more shares is created because when a company distributes dividends which created a shortage and to cover that shortage this company issues further shares or take further financing. This further financing creates Dilution of shares due to which per share book value reduces and as a result, per share market value also goes down.

But if we add or match this reduction in the market value with the value of dividend then it becomes equal meaning that there is not effect of Dilution practically on the shareholder. The other thing which we have discussed in this lecture is that shareholders have an option to adjust their amount of dividend either by selling or by investing their own dividend in the company which we name **Homemade Dividend**.

Module 13: Dividend Policy

Topic 250: Theory of Irrelevance of Dividend-3

In previous lectures, we have talked about the Irrelevance Theory and we have seen that what it is and what does it and what it actually states.

Criticisms on Theory of Irrelevance of Dividend:

But there are many criticisms on this theory and let's see these criticisms one by one among which some points are such that we will further explore in detail.

- **Fractional shares cannot be sold:** It means that if somebody can make Homemade dividend that might require selling of fractional shares meaning that to generate exact amount of shares, it may be that shareholders have to sell fractional shares which may not be possible.
- **Brokerage commission:** When shares are sold on which we have to pay brokerage commission or charges which reduces our return.
- **The percentage holding reduces with the sale of shares:** When we try to make our Homemade dividend by selling some shares from our share holding that reduces the percentage holding of shareholders.
- **Institutions with large holding will not make homemade dividend:** The institutional investors whose percentage holding is more and in some cases it may be the majority holding of a particular institution in a company. If such an institution or mutual funds or a big bank will try to make their homemade dividend by selling their own shares then it will be more affected by it.
- **There are other forces of the market affecting price:** The fluctuation of price in the market is dependent on many other external factors and these external factors do exist then we cannot say that the fluctuation in the market price will only be based on the payment of dividend or retention.
- **Tax rates may be different on dividends and capital gains:** The rate of income tax and the capital gain tax are different and this is again one point that we will discuss more in detail later on.

- **Retained earnings have no cost of financing, whereas others have floatation cost:** It means that retained earnings once the profit is reinvested in the business then it has no cost or zero cost as we have earned profit and invested in our own business but if the company goes for additional financing then it is not free of cost meaning in addition to the return to the shareholders, we have to bear a cost for issuing or floating that financing.
- **Dividend send positive financial signals:** The payment of dividend is something which sends signals meaning that positive or negative signaling also related with dividend.
- **Laws and regulations:** There are laws and regulations which restrict investor to go for homemade dividend and the institutions also.
- **Lack of trust on directors:** The nonpayment of dividend also increases the lack of trust in the directors, the relationship of the investors and the company's directors also affected by it.

Examples:

Let's see some examples which would actually tell us that if we are retaining a part of the earnings or we are distributing the part of earnings.

There may be **three possibilities** for that which are as:

- Our reinvestment rate (as company is reinvesting its profits) is more than the required rate of return of equity then the value of company will go to increase
- And if its required rate of return on equity and reinvestment rate of company is equal then it will be indifferent that company pay or not pay dividend.
- And third possibility may be that if the reinvestment rate of company is less than the required rate of return on equity then the share value of that company will go down.

Example 1: Earnings are all paid as dividend

- Current position: Earnings = \$0.8 per share (all paid out as dividend); RE or required rate of return = 20%, the price per share. would be

- Market value or price = $P = 0.8/0.2 = \$4$ (the PV of constant dividends received in perpetuity).

Example 2: Earnings are reinvested at the cost of equity. Half earnings are retained $RE = 0.2$

- Market price or value = $P = E1(1 - b)/(RE - bR)$
- $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.2) = \4

As half of the earnings or 50% is retained so,

$$\text{Retention (b)} = 50\% = 50/100 = 0.5$$

Where E = earnings

b = retention

RE = cost of equity or return on equity

br = growth rate (where 'b' represent retention and 'r' represent cost of capital)

No change in the share value, and so the dividends are irrelevant.

It means that if the required rate of return on equity and reinvestment rate are equal then there is no difference that dividend is paid or not.

Example 3: earnings are reinvested at more than the cost of equity or if the reinvestment rate is more than the required rate of return of equity

- $RE = 0.2$ (as before) and R (Cost of Capital) = 0.3
- Market price or value = $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.3) = \8

In this case, the share price rises because the extra earnings retained have been invested in a particularly valuable way

Example 4: earnings are reinvested at less than the cost of equity

- $RE = 0.2$ (as before) and R = 0.1
- Market price or value = $P = 0.8(1 - 0.5)/(0.2 - 0.5 \times 0.1) = \2.67

We can conclude our discussion that there is some criticism on the Theory of Irrelevance of Dividend but along with that criticism we have also seen that the only possibility or the only reason for the firm to retain profit should be when they can earn more than what the equity holders require by not giving them dividend.

Module 13: Dividend Policy

Topic 251: Miller and Modigliani's Argument

In this lecture, we will be concluding our discussion on Irrelevance of Dividend or the Theory of Irrelevance.

The M & M (Miller and Modigliani) Dividend Policy:

The M&M Dividend Policy

- The crux of M&M's position is that the effect of dividend payments on shareholder wealth is exactly offset by other means of financing
- The assumptions are certainty and perfect capital markets



Franco Modigliani



Merton Miller

Actually the Theory of Irrelevance was given by Miller and Modigliani and they built-up this argument that why payment of dividend is not relevant to the shareholders and also to the company or the firm. The arguments that we have discussed about the Irrelevance of Dividend most of which we have taken from the theory of Miller and Modigliani.

- The crux of M & M (Miller and Modigliani) theory is that the effect of dividend payments on shareholder wealth is exactly offset by other means of financing. It means that if any company pays dividend to the shareholders and does external financing then its affect vanishes.

- There are two basic things which we need to keep in our mind that the two basic assumptions of this theory or argument are that:
The Certainty or the risk level does not change and also the markets are perfect.

The capital markets being perfect is something which is again not a realistic assumption and second one that is; the risk is also constant is not very realistic either.

Why capital markets are not perfect? What are the perfect markets?

Perfect markets are such capital markets where there are not entry and exit barriers, people have perfect information that is the markets are efficient in accessing the information and people behave rationally or shareholders do everything to maximize their profits.

If we look at the stock or security markets practically, we see that these markets lack all these assumptions that means if the markets will not be perfect certainly the M & M (Miller and Modigliani) argument of Irrelevance of Dividend would not stand correct.

And the second thing is about the risk that is the risk of every firm remains same forever and does external factors not affect it and this is also another assumption which is very little in its depth.

Now, the argument of M & M (Miller and Modigliani) has two aspects:

- The retention of profit does not affect shareholders as they can make homemade dividend. It means that if shareholders do not get dividend or if they get less dividend or if they get more dividend so, they are not worried or concerned about it, it is not relevant to them because if the profit is retained then the share price will go up and as a result they get the benefit. And if they want to convert that benefit into cash then they can sell a part of their shareholding and realize cash.
If the dividend has been paid then their total wealth reduces the market value but their total wealth remains same because if we add the amount of

dividend and the amount of decrease in the value of the share then it becomes same wealth as it can be in the form of nonpayment of dividend meaning market price will remain same.

- When Miller and Modigliani have seen this from other angle then they said that the firm is not affected by the payment of dividend as it can be offset by the external financing. It means that if company pays dividend and obviously it reduces their profits and also cash in the balance sheet which reduces the size of balance sheet also. When the company issues further share to cover this reduction or shortage of cash or financing then they get cash and the size of its balance sheet also becomes equal.

We can conclude our discussion which we will be having in last few lectures regarding the Irrelevance of Dividend that M & M (Miller and Modigliani) argument in favor of or from the shareholder's perspective and also the firm's perspective is simply that the dividends are irrelevant either it is a shareholder or it is a firm but with the assumption that the markets are perfect and the risk level does not change and the two assumptions practically do not exist.

Module 13: Dividend Policy

Topic 252: Issues of Dividend Consideration-1

Before this lecture, we have concluded our discussion on the Relevance and Irrelevance of Dividend. Here we will be talking about a few issues which are generally debated or discussed with reference to Dividend Policy and with reference to the discussion of the Dividend Payout.

Conservation of Value:

The first one issue is Conservation of Value meaning that from any firm's perspective, it is actually the objective of a firm to increase its size, value and to at least conserve the value. Here conserve means to save this value or keep this value as same as its value is now or adopt any way through which this value does not decrease but increases.

So, when we say that **Conservation of Value** then it means that if the firm stands at a certain level or with reference to the market price then it should not go down by adopting a certain dividend policy.

We can say that there will be no effect on the value of the company by increasing or decreasing dividend payout or retention. And if it does then we would say that the value of the firm is not conserved.

Firm cannot create value by altering the dividend payout:

The basic statement is that **Firm cannot create value by altering the dividend payout.** It means that if a firm wants to increase its value for which it increases or decreases the amount of dividend but this thing does not affect the value of this firm.

Investors are able to replicate any dividend stream:

The second thing is that **Investors are able to replicate any dividend stream** that they want. It means that if we want Rs.2 per share dividend then it is not mandatory that company may give this Rs.2 dividend or not but we (as an investor or shareholder) have an option to create this dividend by ourselves. So, the value will be conserved whether we are paying dividend or not.

Lower dividend: sell some shares and get desired cash

If we see both forms that **if any company is paying Lower dividend** or less than their required amount of dividend to any shareholder **then they sell some shares and get desired cash.**

For example, if a shareholder wants to get Rs.5 per share dividend but company has given him Rs.2 per share dividend then by getting Rs.2 per share dividend, the market value or price of his share increases due to this Rs.3 that is the whole argument. And this Rs.3 dividend which has not been given in the form of dividend, then they will get this Rs.3 in the form of that increase in the share price. And shareholders can adjust that amount of dividend to as Rs.5 by selling some shares in the market.

In this way, he has adjusted the lower dividend with higher dividend by selling his shares in the market as from Rs.3 to Rs.5.

Higher dividend: buy additional shares of the company

If the people have more money than their needs then they save them meaning that there are two uses of our income; either it will be expensed out or it will be invested. So, if a person receives more than the expectation, the dividend is more than that suppose we were expecting Rs.5 per share dividend but we have actually got Rs.8 per share dividend then we will adjust this dividend again as; we will keep Rs.5 with us as our requirement and we will put back this remaining Rs.3 in the shares of the company. So, the amount of share is adjusted.

So, in this way, the investors will be able to make their homemade dividend.

We can conclude our discussion that the payment or nonpayment of dividend is not related to the Conservation of Value so, if we are altering the dividend payout or retention ratio, we should not expect that the value of the company will be more or it will be altered by that.

Module 13: Dividend Policy

Topic 253: Issues of Dividend Consideration-2

The second consideration related to the dividend is the **Tax Consideration** and we have seen in previous lectures that this assumption is in both the policies Relevance and Irrelevance Dividend in which we have seen that it is assumed that there is no corporate tax or income tax or individuals don't have to pay taxes.

But in the real world, this is not an assumption which is possible. Practically the corporations, the companies which pay dividend, have to pay taxes similarly the individuals have also not exempted for tax.

So, if tax is one consideration then let's see that how does it actually affects.

Tax Consideration:

- **Personal tax rate on capital gains is different or less than that on dividend income (better not to get dividend).**

It means that there is double taxation on the dividend that when a firm earns some amount of profit then the first step is that the firm pays tax on that and as we know there is one profit which is known as Profit before Tax (PBT) and then income tax is deducted out of that and we get Profit After Tax (PAT). And the second step is that the profit which is remaining after paying the tax is to be distributed among the shareholders and at the time of distribution there should not be any tax because it is just the distribution but practically on the payment of dividend, individuals have to then pay income tax which is the Tax on the Dividend and that is the concept of **Double Taxation.**

So, if this Double Taxation is present then due to its presence we can decide the relevance or irrelevance of dividend with the consideration of tax. And the other thing is that the **capital gain tax** (it is such a tax that when a person sells his/her shares then the tax he/she has to pay is called the capital gain tax) has nothing to do with the profit or loss of the individual meaning

that if an investors has to sell his shares on loss then also he has to pay capital gain tax.

So, considering these points we can understand that tax is a very important consideration.

- **The capital gains tax is deferred until the actual sale of stock (better not to get dividend).**

It means that if an investor gets dividend today the tax will be paid today because dividend has to be paid to the investor after deducting the tax from that and suppose if our country, 10% is the tax rate on dividend and we as an investor has got Rs.100 dividend then company will give us Rs.90 after deducting 10% of Rs.100 or Rs.10 ($\frac{10}{100} \times 100 = 10$) which means that tax on dividend has been paid on that time.

Whereas if it comes to capital gain, the payment of tax is deferred (here **deferred** means that when we will sell our shares and that time we have to pay that tax) so, we can postpone the payment of tax.

- **Paying Capital Gain tax and not Tax on Dividend gives timing option.**

One basic thing that we need to keep in our mind is that any expense/cost/tax that is paid tomorrow is better than if it is paid today. It means that it is good for us as to postpone our payments or costs or taxes or expenses as we can because the money can be reinvested for that period of time which will bring in more profits.

- **Corporate and individual investors have different tax rates.**

Like if a company is investing in other company let's say Mutual Funds is investing in some other company's shares then Mutual Funds have to pay different rate of tax and if individual has to invest in the same company then the tax rate will be different.

- **No impact on tax exempted individuals**

It means that there are some people or groups who are exempted for tax.

If we conclude our discussion about **Tax Consideration** then we can say that there are such many things about tax which we should have to consider and if we will ignore them then we cannot say that dividend policy, payment of dividend is relevant or irrelevant. And all these things are with reference to the payment of tax, tax rates and with reference to the individual and corporations also.

So, this consideration of dividend related issues is particularly extremely important and we need to keep in our mind while we are deciding about the payment or non-payment of dividend.

Module 13: Dividend Policy

Topic 254: Issues of Dividend Consideration-3

In previous lecture, we have talked about few of the considerations related to the Dividend Policy.

Now we will talk about **Floatation Cost** and before we see that what is Floatation Cost, let's first see **why we are talking about Floatation Cost.**

We have seen in the Irrelevance Theory of Dividend that companies can always get more financing if they pay dividend by issuing more shares and this was the concept of Dilution that we have also discussed earlier.

Floatation Cost:

But here there is something which is called the **Floatation Cost of the share** meaning that whenever any company issues shares and that company has to incur a cost for issuing these shares/securities/bonds meaning that the cost which a company has to bear a cost on issuing new shares or securities in the public that is called the Floatation Cost.

Floatation Cost actually includes several other costs which are collectively termed as the Floatation Cost.

Or

The cost associated with issuing securities, such as underwriting, legal, listing and printing fees.

For example, there is a cost of underwriting where the term **underwriting** means that whenever a company has to issue shares, they have to arrange an

underwriter where **underwriter** is such a firm or company whose job is that all these securities or shares must be issued to the public, they should be fully subscribed means that all these securities have to be bought by the public or applied by the public. And if the part of that securities or shares are not subscribed by the general public then the underwriters are responsible for that and they do charge a certain amount for this service of underwriting meaning that they have to pay underwriting fee which is the part of the Floatation Cost.

Similarly when a company issues more capital then there is a **legal cost** involve in that, they have to hire some lawyers or a team of lawyers and a legal team as to incur expenses and hence the legal cost becomes a part of the Floatation Cost.

When the new issued shares are to be listed at the stock market then this would also require more cost and so **listing cost** also becomes the Floatation Cost.

We had the system of physical shareholding where share certificates had issued physically where **printing, stationary cost** has also incurred but at the moment because we have got digital certificates, databases so that the printing and all these costs have been eliminated but nevertheless there is a lot of cost that a company has to bear while issuing new shares or securities which is called the **Floatation Cost**.

Floatation Cost: Formula

$$\text{DGR (Dividend Growth Rate)} = [D/P(1-F)] + g$$

Where:

- D = Dividend
- P = Issue price of share
- F = Floatation cost to Issue price ratio
- g = growth rate

• Floataction Cost: Formula: Example

- Cost of New Equity = $[D/P(1-F)] + g$

- 1. = $[1/10 \times (1 - 7\%)] + 10\%$
= 20.7%

- 2. = $[1/10 \times (1 - 0\%)] + 10\%$
= 20%

We can conclude our discussion that in the consideration of Floataction Cost, we consider some cost which includes underwriting cost, legal cost, listing cost and many other costs or expenses which that company must have to incur while trying to get financing through new securities or issuing new shares.

If Floataction Cost do exist then due to which cost of new equity will change and if this will change then certainly the decision of dividend being irrelevant will not be correct.

Module 13: Dividend Policy

Topic 255: Issues of Dividend Consideration-4

We are having our discussion on the issues of dividend and several considerations that we have talked about and our discussion is still about the considerations related to dividend.

In this lecture, we will talk about two considerations due to which payment of dividend does not remain irrelevant or that makes the payment of dividend relevant. These two are actually the Transaction Cost and Share Divisibility.

Transaction Cost and Share Divisibility:

Transaction Cost means that if a person purchases or sells shares then as a result some expenses are to be incurred.

- **Brokerage fee reduces the return/cash:**

One thing that is called **Income tax** is something different we will not include income tax here because we have already discussed application of income tax separately in detail earlier. And here we are talking about such expense which is called the **Brokerage charges or the Brokerage commission/fee**.

Whenever any shareholder purchases or sells shares then this he has to pay brokerage fee to the broker because in a stock market, we cannot trade without a broker. Every trade buy/sell has to be a broker and the broker's income in the form of commission or fee.

This fee is income for the broker but for the investor it is the expense meaning that if any shareholder will try to make his own homemade dividend but he cannot make exactly the same homemade dividend because whenever he will sell his shares then he has to pay some brokerage fee which reduces the amount of his homemade dividend.

- **Brokerage is paid differently on small and large value stocks:**

Brokerage is something which needs to be understood and a little bit in more detail. For example in a country, the rate of brokerage charges or commission is calculated in two different ways; one way is that a certain percentage of the amount of the trade meaning that how much brokerage we have to pay on the amount of purchase of shares and the second thing is that if we are trading a certain number of shares then how much brokerage we have to pay on per share.

And the actual brokerage is whichever is higher. It means that we have these two options that if we apply one percentage which is given in our rules or we apply a per share cost which is paisa per share then the amount which will be more among them will become our actual brokerage charge.

If we look at two different types of shares; we have got some shares which have got high market value (I.e., Rs.100 or Rs.150 etc.) and the shares which

are called the Penny stocks meaning too less amount shares (I.e., Rs.4 or Rs.10 etc.) then we would see that the brokerage charges are very different. The brokerage charges are be much lesser if we are talking about the high value stocks and the brokerages charges are higher if we are talking about low value stocks.

So, brokerage commission or fee becomes a relevant thing in our decision making regarding the dividend policy.

- **Shares are generally sold in Round Lots:**

The second thing that we need to consider is the **Divisibility of Shares** meaning that if we want to get an exact dividend then it may be that we have to sell shares in fractions (I.e., if we sell 35.5 shares then we will get exact amount of dividend but fractionally shares cannot be sold but there is another interesting thing that is called the **Market Lots and Even Lots or Round Lots** as compared to the **Odd Lots**.

Remember: An **odd lot** refers to an order amount for a security that is less than the normal unit of trading for an asset, which is typically 100 shares for stocks. And In stocks, a **round lot** is considered 100 shares or a larger number that can be evenly divided by 100.

It means that if shares are sold in the form of Market Lots or Round Lots then their prices are more in the market as compare to the Odd Lots meaning that if a share is of Rs.10 then generally its Lot size will be 500 shares which is a **Round Lot**. But if same company sell 250 shares or 350 shares or 195 shares then we name it **Odd Lot** and this is traded in different market which we call **Odd Lot Market** and in these markets, the price of these share is less than the regular price.

So, as a result we can understand that if we are talking about the **Irrelevance of Dividend** then we must consider two things meaning that it is crux of today's overall discussion that **we must have to consider the Transaction Cost or Divisibility of Shares** and if we will not consider that then we will do a large mistake with reference to the Dividend Policy because the shareholders cannot make their homemade dividend till they are allowed to sell fractional shares and whenever the market value of Odd Lot and Round Lot is equal which may not be equal.

And the other thing is that if there will be brokerage fee then it will certainly affect the dividend or homemade dividend.

So, we can conclude our discussion that the two elements which are Transaction Cost and Divisibility of Shares needs to be very seriously considered because it is not possible to split shares and the Odd Lot and Round Lot prices are not the same and if these two prices are not equal then we cannot say that the homemade dividend and the company's dividend can be treated in the same way.

Brokerage fee is one big reality and it is not possible to have any trade without the brokerage fee. Therefore brokerage fees or Divisibility of Shares are these to extremely important considerations which make the dividends relevant and negate the Irrelevance Theory.

Module 13: Dividend Policy

Topic 256: Issues of Dividend Consideration-5

This is the last of our discussion on the considerations related to the dividend policy and here we will talk about few things.

Institutional Restrictions:

The first thing is Institutional Restrictions and **Institution** means that those investors which invest in the form of large companies or firms i.e., commercial or investment banks which invest money in stock or share markets, Brokerage houses or Mutual Funds also invest their money in stock market.

These institutions actually have a very set investment policy and in that investment policy, there are certain restrictions because of which they cannot behave like individual investors. It means that if a person is an individual investor then it is his will that whenever he/she purchase or sell his/her shares, try to make homemade dividend, hold for dividend or not, all these possibilities are present. But when we look at an investment from an Institution's perspective then there are certain restrictions and which make the dividend policy relevant. These restrictions are given below:

Companies which pay regular dividend:

In many of the cases, institutions would like the funds to be invested in the companies which pay regular dividend meaning that the companies which do not pay dividend then there is or may be an Institutional Restriction that do not invest in such companies. So, that will make the payment of dividend a relevant.

Investments from Endowments are restricted for Capital Gains:

Similarly, there are investments out of Endowment funds and **Endowment fund** is a big amount which has to be invested for a specific purpose on which coming income or return will be used for that purpose. For example, if there is an Education Endowment Fund then this means that whatever interest or profit or return or dividend that will be earned from that Education Endowment Fund will be used for the education of children or students.

So, Endowment Funds would not allow the care takers or managers of the fund to go for capital gains.

Restriction on Liquidation of Capital:

And lastly the liquidation of capital is also be restricted, it is possible that it is restricted in case of institutions meaning that if we liquidate our investment today then it may be that this value may not be the same in the future as it is today and liquidation time period may also be set by the institutions that we may not liquidate till 2 or 4 or 7 years.

Financial Signaling:

Signal means that something which tells us about what is happening. When we talk about dividends, it is said that dividends actually send some signals, these signals are to the investors and to the stakeholders.

- Financial signaling is very much important and relevant when the markets are imperfect or we can say that **financial signaling is based on market imperfections.**
- **Dividends send signals about a firm's profitability.** It means that dividend actually show Company's potential or strength of earning profit and

dividend is not paid only out of the current profit but it can be paid out of the retained profits of the previous years.

So, if a company is paying regular dividend and even in a year when the company has not earned any profit but paid dividend to its investor or shareholders from its retained earnings then that sends a very strong signal to the investors that it is a very stable company and financially this company is not going to go down.

- **Dividends are better than announcements of profit/growth.** When a company announces that how much income we have earned in a particular year, this is one signal but it is said that payment of dividend is a better or stronger signal about the strength of the company than just announcing that how much earnings we have in this years.
- **Consistent dividend payment signals; better than what it looks.** It means that if any investor gets regular consistent dividend then this stability of dividend is perceived good and prioritize and shareholders want that they get this dividend also.
- **Share price responds to signals.** It means that the price of the company's share is also affected by the payment of dividend as any company declares dividend and we see that this company's share price starts to go up so, we can say that the market response to the news of dividend. Although it happens that once the dividend is paid, the share price go down because of the reduction in the value but then it is something which sends a very strong signal.

We can conclude our discussion that the two considerations which need to be accounted for are; the financial signaling effect which is through the payment of dividend that should always be considered if a company wants to have a strong financial image in the eyes of the investor. And second this is that there is a large amount of investment in the shares of every company made by the institutions and institutions are very much restricted.

So, considering these two above points again we can say that the payment of dividend is relevant.